

QUESTION 1

SOURCE CODE:

```
#include <iostream>
using namespace std;

void swap(int *num1, int *num2){
    int temp;
    temp=*num1;
    *num1=*num2;
    *num2=temp;
}

void mean(int a, int b, float *mean_val){
    *mean_val=(a+b)/2.0;
}

void range(int a, int b,int *range){
    if(a>b){
        *range = (a-b);
    }
    else{
        *range = (b-a);
    }
}

void normalized(int a, int b, int *norm_a, int *norm_b){
    *norm_a=(a)/(a);
    *norm_b=(b)/(b);
}
```

```

int main(){
    int num1=10;
    int num2=20;

    cout<<"num1 before swap " <<num1<<endl;
    cout<<"num2 before swap " <<num2<<endl;
    int range_val=0;
    float mean_val=0;
    int norm_a=0;
    int norm_b=0;

    swap(&num1, &num2);
    cout<<"num1 after swap " <<num1<<endl;
    cout<<"num2 after swap " <<num2<<endl;

    mean(num1,num2,&mean_val);
    cout<<"mean " <<mean_val<<endl;

    range(num1,num2,&range_val);
    cout<<"range " <<range_val<<endl;

    normalized(num1,num2,&norm_a,&norm_b);
    cout<<"Normalized value of first number " <<norm_a<<endl;
    cout<<"Normalized value of second number " <<norm_b<<endl;

    return 0;
}

```

OUTPUT:

```

num1 before swap 10
num2 before swap 20
num1 after swap 20
num2 after swap 10
mean 15
range 10
Normalized value of first number 1
Normalized value of second number 1

```

QUESTION 2

SOURCE CODE:

```
#include <iostream>
using namespace std;
#define SUB 3
void tot_marks(int *marks, int *sum){
    *sum=0;
    for(int i=0;i<SUB;i++){
        *sum += *(marks+i);
    }
}
void percentage(int *sum, float *percent){
    *percent = (*sum/ (SUB*100.0))*100;
}
int main(){
    int sub_marks[SUB];

    for(int i=0;i<SUB;i++){
        cout<<"MARKS OF SUB "<<i+1<<endl;
        cin>>sub_marks[i];
    }
    int sum;
    float percent;
    tot_marks(sub_marks,&sum);
    percentage(&sum,&percent);

    cout<<"Sum = "<<sum<<endl;
    cout<<"Percent = "<<percent;
    return 0;
}
```

OUTPUT:

```
MARKS OF SUB 1
56
MARKS OF SUB 2
80
MARKS OF SUB 3
76
Sum = 212
Percent = 70.6667
c:\Sem 2\OOP LABS>
```